Preregistration for:

Quantifying the Distortive Impacts of LLM Writing Assistance

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Submission Date: 10th of June 2025

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1 Study Information

1.1 Title

Quantifying the Distortive Impacts of LLM Writing Assistance

1.2 Authors

Paul Röttger, Kobi Hackenburg, Hannah Rose Kirk, Dirk Hovy, and Christopher Summerfield.

1.3 Description

Our study investigates whether the use of Large Language Models (LLMs) for writing assistance systematically distorts how others perceive LLM users and the opinions they express.

The study employs a two-phase design:

In **Phase 1 (Writing)**, participants ("writers") report their demographic characteristics and then respond to political propositions by providing numeric stance ratings, opinion bullet points and opinion paragraphs. Writers also report their emotional states while writing, their issue knowledge and how important each issue is to them. For each proposition, each writer is then randomly assigned to one of three LLMs, which generates a version of the writer's opinion paragraph based on one of three randomly assigned input conditions. Writers can edit these AI-generated paragraphs and select which version they prefer for communicating their views.

In **Phase 2 (Rating)**, participants ("raters"), who are distinct from the Phase 1 writers, evaluate the paragraphs produced in Phase 1, without knowing the provenance of the paragraphs. Raters assess paragraph attributes (e.g. originality), make inferences about writer attributes (e.g. issue knowledge), writer personality traits (e.g. optimism), as well as writer demographics and emotional states.

This design allows for measuring systematic differences in how human-written versus AI-assisted content is perceived, quantifying potential distortive effects of LLM writing assistance.

1.4 Research Questions

Our two main research questions are:

RQ1 [Extent of Perceived Distortion]: To what extent does the use of LLMs for writing assistance systematically distort how others perceive 1) LLM users and 2) the opinions they express?

RQ2 [Variation in Perceived Distortion]: How do these perceptions vary depending on 1) which LLM is used for writing assistance and 2) what type of input is provided to the LLM?

A third, auxiliary research question concerns the ecological validity of our setup:

RQ3 [Writer Preference for LLM Writing]: To what extent do writers prefer content written with LLM assistance to their own independent writing, and how does this preference vary by choice of LLM and the type of input provided to the LLM?

A fourth, auxiliary research question concerns the first-person desirability of distortion:

RQ4 [Writer Preference for Distortion]: To what extent do the distortive effects of LLM writing assistance align with the preferences for distortion expressed by the writers using the LLMs?

2 Design Plan

2.1 Study Type

Experiment: A researcher randomly assigns treatments to study subjects. This includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.

2.2 Blinding

The human subjects in this study will not know the treatment group to which they have been assigned.

2.3 Study Design

Our study will use a two-phase design to investigate the distortive impacts of LLM writing assistance.

2.3.1 Onboarding Questionnaire

All participants will complete an onboarding questionnaire before starting the study. The onboarding questionnaire will ask participants about their age, gender, race, English language competency, highest level of education, current yearly income, political party preference, and political ideology.

2.3.2 Phase 1: Writing

In Phase 1, participants ("writers") will engage with political propositions and generate opinion content. For each writer, the process will be as follows:

1. The writer is randomly assigned to 3 of 100 political propositions (see §2.3.4)
2. For each proposition, the writer reports:
   (a) Their level of agreement with the proposition (0-100 scale)
   (b) Key reasons for their opinion (2-4 bullet points)
   (c) A paragraph explaining their opinion (100-150 words)
   (d) Emotional state while writing their paragraph, measured on an affect grid (Russell et al., 1989)
   (e) Their knowledge of the issue (0-100 scale)
   (f) The importance of the issue to them (0-100 scale)
   (g) Confirmation of their level of agreement with the proposition (0-100 scale)
   (h) Confidence in their opinion (0-100 scale)
3. For each writer, we then create a balanced assignment where:
   (a) Each writer engages with all 3 LLMs (listed in §2.3.5), with one LLM assigned to each proposition
   (b) Each writer experiences all 3 input sub-conditions (listed in §2.3.6), with one sub-condition assigned to each proposition
4. For each proposition/paragraph, the writer:
   (a) Rates the perceived stance of the LLM-generated paragraph (0-100 scale)
   (b) Edits the LLM-generated paragraph to match their own opinion (if desired)
   (c) Chooses between their original and the edited paragraph for communicating their opinion
   (d) Provides reasons for their preference
   (e) Confirms again their level of agreement with the proposition (0-100 scale)

5. After engaging with all propositions, writers complete a questionnaire about which types of distortion from AI writing assistance they would find desirable or undesirable. The types of distortion they are asked about (e.g. AI making them seem more/less confident) match the rating attributes in Phase 2.

For each writer and each proposition, this will result in three paragraphs: one human-written, one LLM-written, and one co-written (human-edited) paragraph. Note that, since editing is optional, the co-written paragraph may be identical to the LLM-written paragraph. In such cases, we will carry forward to Phase 2 only the LLM-written paragraphs.

2.3.3 Phase 2: Rating

In Phase 2, independent participants ("raters") will evaluate the paragraphs produced in Phase 1, without knowing whether these paragraphs were human-written, LLM-written, or co-written. For each rater, the process will be as follows:

1. The rater is randomly assigned to 10 paragraphs from Phase 1. Assignment is such that every paragraph receives at least 10 independent ratings.
2. For each paragraph, the rater provides several categories of ratings:
   (a) **Inferred issue-specific writer attributes:**
      - Writer agreement with the proposition (0-100 scale)
      - Writer confidence in their opinion (0-100 scale)
      - Writer knowledge about the issue (0-100 scale)
      - Importance of the issue to the writer (0-100 scale)

   (b) **Paragraph-level attributes:**
      - Relevance/on-topic nature (0-100 scale)
      - Clarity of writer's opinion (0-100 scale)
      - Formality of language (0-100 scale)
      - Informativeness/educational value (0-100 scale)
      - Originality/novelty of arguments (0-100 scale)

   (c) **Inferred writer personality and values:**
      - Friendliness (0-100 scale)
      - Optimism (0-100 scale)
      - Collectivism (0-100 scale)
      - Openness to reconsidering views (0-100 scale)

   (d) **Perceived paragraph emotions:**
      - Hope (0-100 scale)
      - Excitement (0-100 scale)
      - Fear (0-100 scale)
      - Disgust (0-100 scale)
      - Anger (0-100 scale)

   (e) **Inferred writer emotional state:**
      - Location on affect grid (Russell et al., 1989)

(f) **Inferred writer demographics:**

- Age (numeric response)
- Gender (categorical response)
- Race/ethnicity (categorical response)
- English as first language (categorical response)
- English language skills (categorical response)
- Education level (categorical response)
- Income level (categorical response)
- Political party preference (categorical response)
- Political ideology (categorical response)

2.3.4 Political Propositions

We will include 100 political propositions in our study. We constructed these propositions based on the 212 political issues collected for IssueBench (Röttger et al., 2025), using the following process:

1. We removed out-of-scope issues based on IssueBench metadata:
   (a) We excluded issues related to criminal activity and hateful ideology.
   (b) We excluded issues related to non-UK-relevant specific geographies.
   (c) We excluded issues concerning historical events before 2020.
2. For each remaining issue (e.g. "the COVID-19 vaccine"), we used GPT-4o to generate a UK-relevant proposition that reflects a clear stance (e.g. "The UK should mandate COVID-19 vaccinations for all eligible citizens.").
3. We removed propositions that are duplicates, not about specific issues, or non-UK-applicable.
4. We edited the remaining propositions to ensure specificity, realism, and UK relevance.

Many of the resulting propositions clearly align with left or right political ideology. We edit and then select 100 propositions such that the overall set of propositions is balanced in its political leaning based on manual assessment. The full list of propositions will be made available in the online supplementary materials.

2.3.5 Selected LLMs

We will include 3 LLMs in our study, which we select to broadly represent the state-of-the-art in June 2025 and to cover diverse LLM development contexts.

1. GPT-4o, developed by OpenAI
2. Claude Sonnet 4, developed by Anthropic
3. DeepSeek V3, developed by DeepSeek

We will use OpenRouter for all model inference. Model temperature will be set to 1, a common default value, to match natural user interaction.

2.3.6 LLM Input Sub-Conditions

We will include 3 LLM input sub-conditions in our study. These input sub-conditions, shown in Table 1, will determine, for a given writer/proposition pair, what inputs from the writer are used to generate the LLM-written paragraph in Phase 1.

Note that, for the paragraph-based condition, we randomly assign either a "rewrite" or "improve" variant with equal probability. Our expectation us that these two variants this will not produce meaningfully different outputs or writer perceptions. If significant differences emerge, we will report these separately and consider including this distinction as an additional factor in our models.

Condition	**Prompt Template**
1: Stance-based	Consider the following political proposition: [proposition] On a scale from 0 to 100, where 0 is "strongly disagree", 50 is "ambivalent", and 100 is "strongly agree", my opinion on this proposition is [writer_stance]. Please write a short paragraph of 100-150 words for me which explains my opinion. Do not mention the numeric rating or include any preamble like "Based on the rating, ...". Reply only with the paragraph, nothing else.
2: Bullet-based	Consider the following political proposition: [proposition] My opinion on this proposition is described by the following bullet points: [writer_bullets] Please write a short paragraph of 100-150 words for me which explains my opinion. Do not include any preamble, like "Based on the bullet points ...". Reply only with the paragraph, nothing else.
3: Paragraph-based	Consider the following political proposition: [proposition] I wrote the following paragraph to explain my opinion on this proposition: "[writer_paragraph]" Please {rewrite|improve} this paragraph without changing its length. Do not include any preamble, like "Based on the original paragraph ...". Reply only with the {rewritten|improved} paragraph, nothing else.

Table 1: The **3 LLM input sub-conditions** that we will use in our study. Each condition corresponds to a prompt template, which we will use to prompt LLMs to generate LLM-written paragraphs. For condition 3, we randomly select either "rewrite" or "improve" with equal probability.

2.4 Randomisation

For each writer, we create a balanced assignment where:

1. Each writer engages with all 3 LLMs (listed in §2.3.5), with one LLM assigned to each proposition.
2. Each writer experiences all 3 input sub-conditions, with one sub-condition assigned to each proposition.

The 3 LLMs and the 3 input sub-conditions are randomly paired with each other. This balanced assignment ensures that each writer produces data points for all three LLMs and all three sub-conditions, allowing for more robust comparisons between models and input types.

3 Sampling Plan

3.1 Existing Data

Registration prior to creation of data: As of the date of submission of this research plan for preregistration, the data have not yet been collected, created, or realised.

3.2 Data Collection Procedures

Data for both Phases will be collected using custom web-based survey interfaces, implementing the procedures described in the Study Design section. Pasting text will be disabled in the Phase 1 survey interface to reduce the likelihood of writers using AI to complete the writing portions of the survey.

We will recruit all participants through Prolific. The participant pool will be restricted to Prolific users aged 18 and over whose current country of residence is the United Kingdom. For Phase 1, we will recruit a representative sample using the representative participant pool provided by Prolific. For Phase 2, which

involves a much larger number of participants, we will not require a representative sample. In the unlikely case that we encounter difficulties recruiting enough participants via Prolific, we will recruit the remainder via another crowd-sourcing platform whilst maintaining the same eligibility criteria. Payment will be at a rate of £9 per hour, based on median completion time, as recommended by Prolific.

3.3 Sample Size

We will recruit ∼1,500 participants ("writers") for Phase 1 and ∼12,000 participants ("raters") for Phase 2.

For Phase 1, with 1,500 writers each assigned to 3 propositions, we will collect ∼4,500 writer paragraphs. Each writer will engage with 1 of 3 LLMs and 1 of 3 input sub-conditions per proposition, resulting in ∼500 paragraphs for each of the 9 LLM-input condition combinations. For each writer paragraph, there will be one LLM-written paragraph and one co-written paragraph.

For Phase 2, with each paragraph receiving at least 10 independent ratings, and each rater rating 10 paragraphs, we will need up to 13,500 raters to evaluate all paragraphs generated in Phase 1. We expect the actual number to be lower, since we expect that at least some writers will choose not to edit the LLM-written paragraphs (§2.3.2), thus reducing the number of distinct co-written paragraphs that require rating.

The exact number of participants will be determined based on budget constraints, and depending on actual study completion time for participants in each Phase.

3.4 Sample Size Rationale

Our sample size is primarily determined by:

1. Our factorial design, with 3 LLMs and 3 input sub-conditions, requiring sufficient coverage
2. The requirement for multiple independent ratings per paragraph to measure rater agreement and reduce the impact of individual rater bias
3. Budget constraints for participant compensation

While we cannot expect high statistical power for individual paragraphs or propositions, our design allows for robust analysis across experimental conditions. The large number of ratings will enable us to detect systematic differences in perception between human-written, LLM-written, and co-written content, as well as between content written by different LLMs.

3.5 Stopping Rule

Data collection will proceed until we reach our target sample sizes for each Phase or exhaust our allocated budget. We will not analyse the data until collection is complete for each Phase.

4 Variables

4.1 Measured Variables: Onboarding

Prior to the experimental portions of the study, we will collect data on several participant attributes in an onboarding questionnaire. The exact question wordings are detailed below.

Age: What is your age?
[Numeric response]

Gender: What is your gender?
[Male / Female / Other / Prefer not to say]

Race/Ethnicity: What is your race / ethnicity?
[White / Black / Asian / Mixed / Other / Prefer not to say]

English Language 1: Is English your first language?
[Yes / No / Bilingual from birth / Prefer not to say]

English Language 2: How would you rate your English language skills?
[Basic (can understand simple content with difficulty) / Intermediate (can understand and communicate everyday topics) / Advanced (can communicate complex ideas effectively) / Expert (can use English fluently for all purposes including professional writing) / Prefer not to say]

Education: What is the highest level of education you have completed?
[GCSEs or equivalent / A-levels or equivalent / Vocational qualification / Undergraduate degree / Postgraduate degree (Master's) / Doctorate (PhD) / Other / Prefer not to say]

Income: What is your current yearly income?
[Under £15,000 / £15,000-£24,999 / £25,000-£34,999 / £35,000-£49,999 / £50,000-£74,999 / £75,000-£99,999 / £100,000+ / Prefer not to say]

Political Party: Which political party did you vote for in the 2024 UK general election?
[Conservative / Labour / Liberal Democrats / Scottish National Party / Greens / Other / Did not vote / Not eligible to vote / Prefer not to say]

Political Ideology: How would you describe your political ideology?
[Very Left-Wing / Moderately Left-Wing / Centrist / Moderately Right-Wing / Very Right-Wing / Other / Prefer not to say]

4.2 Measured Variables: Phase 1

The variables we measure in Phase 1 of our study capture the writers' original opinions, their perceptions of LLM-written content, and their preferences between human-written versus co-written content.

4.2.1 Step 1: Writer's Original Opinion

First, we ask the following questions for each proposition. These questions are asked across separate screens, where the proposition is shown at the top of each screen ("Consider this statement: [proposition]").

Proposition Agreement: What is your level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Opinion Bullets: Please explain the key reasons for your opinion in 2-4 bullet points. Start each bullet point with a dash - and put each bullet point on a separate line.
[Free text response]

Opinion Paragraph: Please expand your bullet points into a short paragraph (around 100-150 words), in which you explain your opinion.
[Free text response]

Affect: We would like to understand how you felt while writing your paragraph. Please tell us by clicking on the grid below:
[Affect grid visual with pleasure-displeasure and arousal-sleepiness dimensions]

Issue Knowledge: How much do you know about the issue referenced in the statement?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

Issue Importance: How important is the issue referenced in the statement to you?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

Proposition Agreement 2: Please tell us again: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Opinion Confidence: How confident are you in your opinion on this issue?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

4.2.2 Step 2: Engagement with LLM-Generated Content

After completing Step 1 for each proposition, we ask writers to engage with content generated by LLMs based on their inputs from Step 1. These questions are also asked across separate screens, where the proposition is shown at the top of each screen ("Consider this statement: [proposition]"). We also introduce the LLM-written paragraph ("Here is another paragraph, responding to the same statement you replied to. This paragraph was written by AI based on what you wrote earlier: [model_paragraph]").

Perceived Stance: What is this paragraph's level of agreement with the statement?
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Edited Paragraph: Please edit the AI-written paragraph until you feel that it reflects your opinion about the statement. If you feel that the paragraph already reflects your opinion, you do not have to edit it. Otherwise please edit it until it does.
[Free text response, starting with the model-generated paragraph]

Paragraph Preference: Imagine you want to explain your opinion on the statement to someone. Which of the two paragraphs below would you rather use for this purpose?
[The paragraph on the left (original) / Equal preference / The paragraph on the right (edited)]

Preference Reason: [If answer is not "equal preference"] Why do you prefer this paragraph over the other? Please tick all reasons that apply and explain any additional reasons under "other".
[Checkbox options: "It better reflects my opinion." / "I prefer the writing style." / "Other (please specify):"]

Proposition Agreement 3: Please tell us one last time: What is your level of agreement with the statement? By default, the slider is set to the answer you gave before. If your opinion has changed, please move the slider accordingly.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

4.2.3 Step 3: Distortion Acceptability

After completing Steps 1 and 2 for all propositions, we ask writers about which types of distortion from using AI writing assistance they would find desirable or undesirable. The types of distortion they are asked about directly correspond to the rating attributes in Phase 2. Each question is asked on a separate screen, where the top of the screen reads "When using AI for writing assistance, the resulting text might be perceived differently from how your own independent writing (without AI assistance) would be perceived. We want to understand which differences you would find desirable or undesirable.". All responses are measured on a 0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree".

The exact questions are:

1. "I am okay with AI writing assistance making my opinion on an issue seem more extreme than my own independent writing would make it seem."
2. "I am okay with AI writing assistance making my opinion on an issue seem more moderate / balanced than my own independent writing would make it seem."
3. "I am okay with AI writing assistance making me seem more confident in my opinion than my own independent writing would make me seem."
4. "I am okay with AI writing assistance making me seem less confident in my opinion than my own independent writing would make me seem."
5. "I am okay with AI writing assistance making me seem more knowledgeable about an issue than my own independent writing would make me seem."
6. "I am okay with AI writing assistance making me seem less knowledgeable about an issue than my own independent writing would make me seem."
7. "I am okay with AI writing assistance making an issue seem more important to me than my own independent writing would make it seem."
8. "I am okay with AI writing assistance making an issue seem less important to me than my own independent writing would make it seem."
9. "I am okay with AI writing assistance making my writing seem more relevant or on-topic."
10. "I am okay with AI writing assistance making my writing seem less relevant or on-topic."
11. "I am okay with AI writing assistance making my opinion clearer to others than my own independent writing would make it."
12. "I am okay with AI writing assistance making my opinion less clear to others than my own independent writing would make it."
13. "I am okay with AI writing assistance making my writing seem more formal."
14. "I am okay with AI writing assistance making my writing seem less formal."

15. "I am okay with AI writing assistance making my writing seem more informative or educational."
16. "I am okay with AI writing assistance making my writing seem less informative or educational."
17. "I am okay with AI writing assistance making my writing seem more original or novel."
18. "I am okay with AI writing assistance making my writing seem less original or novel."
19. "I am okay with AI writing assistance making me seem more friendly than my own independent writing would make me seem."
20. "I am okay with AI writing assistance making me seem less friendly than my own independent writing would make me seem."
21. "I am okay with AI writing assistance making me seem more optimistic than my own independent writing would make me seem."
22. "I am okay with AI writing assistance making me seem less optimistic than my own independent writing would make me seem."
23. "I am okay with AI writing assistance making me seem more community-oriented than my own independent writing would make me seem."
24. "I am okay with AI writing assistance making me seem less community-oriented than my own independent writing would make me seem."
25. "I am okay with AI writing assistance making me seem more open to reconsidering my views than my own independent writing would make me seem."
26. "I am okay with AI writing assistance making me seem less open to reconsidering my views than my own independent writing would make me seem."
27. "I am okay with AI writing assistance making my writing express more hope, as perceived by others."
28. "I am okay with AI writing assistance making my writing express less hope, as perceived by others."
29. "I am okay with AI writing assistance making my writing express more excitement, as perceived by others."
30. "I am okay with AI writing assistance making my writing express less excitement, as perceived by others."
31. "I am okay with AI writing assistance making my writing express more fear, as perceived by others."
32. "I am okay with AI writing assistance making my writing express less fear, as perceived by others."
33. "I am okay with AI writing assistance making my writing express more disgust, as perceived by others."
34. "I am okay with AI writing assistance making my writing express less disgust, as perceived by others."
35. "I am okay with AI writing assistance making my writing express more anger, as perceived by others."
36. "I am okay with AI writing assistance making my writing express less anger, as perceived by others."
37. "I am okay with AI writing assistance making me seem older than what people would guess if they had read my own independent writing."
38. "I am okay with AI writing assistance making me seem younger than what people would guess if they had read my own independent writing."
39. "I am okay with AI writing assistance making my gender seem different than what people would guess if they had read my own independent writing."
40. "I am okay with AI writing assistance making my race or ethnicity seem different than what people would guess if they had read my own independent writing."

41. "I am okay with AI writing assistance making my English language skills seem better than what people would guess if they had read my own independent writing."
42. "I am okay with AI writing assistance making my English language skills seem worse than what people would guess if they had read my own independent writing."
43. "I am okay with AI writing assistance making me seem more educated than what people would guess if they had read my own independent writing."
44. "I am okay with AI writing assistance making me seem less educated than what people would guess if they had read my own independent writing."
45. "I am okay with AI writing assistance making my income seem higher than what people would guess if they had read my own independent writing."
46. "I am okay with AI writing assistance making my income seem lower than what people would guess if they had read my own independent writing."
47. "I am okay with AI writing assistance making it seem like I voted for a different political party than what people would guess if they had read my own independent writing."
48. "I am okay with AI writing assistance making me seem more politically extreme than my own independent writing would make me seem."
49. "I am okay with AI writing assistance making me seem more politically moderate / centrist than my own independent writing would make me seem."

4.3 Measured Variables: Phase 2

Our primary outcome variables are the perceptions measured in Phase 2 of the study, where raters evaluate paragraphs produced in Phase 1. For each paragraph, raters complete a series of rating questions.

4.3.1 Inferred Issue-Specific Writer Attributes

Writer Agreement: What is the author's level of agreement with the statement? Please read the text carefully.
[0-100 scale, where 0 = "Strongly disagree", 50 = "Ambivalent", and 100 = "Strongly agree"]

Writer Confidence: How confident is the author of the paragraph in their opinion?
[0-100 scale, where 0 = "Not confident at all", 50 = "Moderately confident", and 100 = "Extremely confident"]

Writer Knowledge: How much does the author of the paragraph know about this issue?
[0-100 scale, where 0 = "No knowledge at all", 50 = "Some understanding", and 100 = "Expert knowledge"]

Issue Importance: How important is this issue to the author of the paragraph?
[0-100 scale, where 0 = "Not important at all", 50 = "Moderately important", and 100 = "Extremely important"]

4.3.2 Paragraph-Level Attributes

Relevance: How on-topic / relevant is the paragraph to the statement?
[0-100 scale, where 0 = "Not relevant at all", 50 = "Moderately relevant", and 100 = "Extremely relevant"]

Clarity: How clear is the author's opinion in the paragraph?
[0-100 scale, where 0 = "Not clear at all", 50 = "Moderately clear", and 100 = "Extremely clear"]

Formality: How formal is the language used in the paragraph?
[0-100 scale, where 0 = "Not formal at all", 50 = "Moderately formal", and 100 = "Extremely formal"]

Informativeness: How informative / educational did you find the paragraph?
[0-100 scale, where 0 = "Not informative at all", 50 = "Moderately informative", and 100 = "Extremely informative"]

Originality: How original / surprising / novel did you find the arguments in the paragraph?
[0-100 scale, where 0 = "Not original at all", 50 = "Moderately original", and 100 = "Extremely original"]

4.3.3 Inferred Writer Personality and Values

Friendliness: How friendly is the author of the paragraph?
[0-100 scale, where 0 = "Extremely unfriendly" (Confrontational, Hostile, Argumentative), 50 = "Neither friendly nor unfriendly", and 100 = "Extremely friendly" (Warm, Welcoming, Supportive)]

Optimism: How optimistic is the author of the paragraph?
[0-100 scale, where 0 = "Highly pessimistic" (Focuses on problems, Anticipates negative outcomes, Sees limitations), 50 = "Balanced outlook", and 100 = "Highly optimistic" (Emphasises possibilities, Expects positive outcomes, Sees opportunities in challenges)]

Community Orientation: How community-oriented is the author of the paragraph?
[0-100 scale, where 0 = "Not community-oriented" (Prioritises personal autonomy, Emphasises individual responsibility, Values self-reliance), 50 = "Balanced between individual and community concerns", and 100 = "Highly community-oriented" (Prioritises community welfare, Emphasises shared responsibility, Values interdependence)]

Openness: How open is the author of the paragraph to reconsidering their views?
[0-100 scale, where 0 = "Firmly committed to current views" (Resistant to counterarguments, Confident in existing position, Values consistency), 50 = "Moderately open to reconsideration", and 100 = "Highly open to re-

considering views" (Welcomes different perspectives, Weighs new evidence, Willing to change position)]

4.3.4 Perceived Paragraph Emotions

Hope: Does the paragraph express hope?
[0-100 scale, where 0 = "No sign of hope", 50 = "Moderate hope", and 100 = "Extreme hope"]

Excitement: Does the paragraph express excitement?
[0-100 scale, where 0 = "No sign of excitement", 50 = "Moderate excitement", and 100 = "Extreme excitement"]

Fear: Does the paragraph express fear?
[0-100 scale, where 0 = "No sign of fear", 50 = "Moderate fear", and 100 = "Extreme fear"]

Disgust: Does the paragraph express disgust?
[0-100 scale, where 0 = "No sign of disgust", 50 = "Moderate disgust", and 100 = "Extreme disgust"]

Anger: Does the paragraph express anger?
[0-100 scale, where 0 = "No sign of anger", 50 = "Moderate anger", and 100 = "Extreme anger"]

4.3.5 Inferred Writer Emotional State

Affect Grid: How did the author feel while writing the paragraph? Please tell us by clicking on the grid below.
[Affect grid visual with pleasure-displeasure and arousal-sleepiness dimensions]

4.3.6 Inferred Writer Demographics

Age: What is the author's age?
[Numeric response]

Gender: What is the author's gender?
[Male / Female / Other]

Race/Ethnicity: What is the author's race / ethnicity?
[White / Black / Asian / Mixed / Other]

English Language 1: Is English the author's first language?
[Yes / No]

English Language 2: How would you rate the author's English language skills?
[Basic (can understand simple content with difficulty) / Intermediate (can understand and communicate everyday topics) / Advanced (can communicate complex ideas effectively) / Expert (can use English fluently for all purposes including professional writing)]

Education: What is the highest level of education that the author has completed?
[GCSEs or equivalent / A-levels or equivalent / Vocational qualification / Undergraduate degree / Postgraduate degree (Master's) / Doctorate (PhD) / Other]

Income: What is the author's current yearly income?
[Under £15,000 / £15,000-£24,999 / £25,000-£34,999 / £35,000-£49,999 / £50,000-£74,999 / £75,000-£99,999 / £100,000+]

Political Party: Which political party did the author vote for in the 2024 UK general election?
[Conservative / Labour / Liberal Democrats / Scottish National Party / Greens / Other / Did not vote / Not eligible to vote]

Political Ideology: What is the author's political ideology?
[Very Left-Wing / Moderately Left-Wing / Centrist / Moderately Right-Wing / Very Right-Wing / Other]

4.4 Attention Checks

We will implement different screening mechanisms for each Phase of the study to ensure data quality.

For Phase 1 (Writing), we will include a simple writing screener asking participants to name their favourite ice cream flavour and explain their choice in 1-2 sentences. This screener will be assessed by GPT-4o, where any on-topic answer will be considered a pass. The screener will be shown just after the onboarding questionnaire. Participants who fail this screener will be excluded from the study and receive partial compensation.

For Phase 2 (Rating), we will present participants with a scale that ostensibly measures how convincing they find different types of writing, but will include clear instructions to ignore the question about convincing writing and instead move the slider to a specific range (between 30 and 40). The screener will be shown just after the onboarding questionnaire. Participants who fail this attention check will be excluded from the study but will receive partial compensation.

5 Analysis Plan

5.1 RQ1: Extent of of Third-Person Perceived Distortion

Our first main research question is concerned with establishing to what extent the use of LLM writing assistance systematically distorts how others perceive LLM users and their opinions.

For each paragraph and each rating attribute, we will have collected sets of ratings from at least 10 independent raters during Phase 2 of our study. We expect variation in ratings across annotators for all rating attributes, including noise from inattentive raters. We will measure the extent of inter-annotator variation using standard deviation (for continuous variables) as well as simple percentage agreement and Krippendorff's Alpha (for categorical variables).

To answer RQ1, we will run separate analyses for each rating attribute. Each observation will be a paragraph-rater pair, i.e. the rating given by a specific rater for a specific paragraph. For each rating attribute, we will fit **mixed-effects regression models**, specified as follows:

$$\begin{aligned}\text{attribute} \sim\ & \text{paragraph_type}+ \\ & (1 \mid \text{proposition_id})+ \\ & (1 \mid \text{writer_id})+ \\ & (1 \mid \text{rater_id})\end{aligned}$$

where:

- **attribute** is the rating attribute of interest.
- **paragraph_type** is a nominal categorical variable taking 3 values: writer_paragraph, model_paragraph, edited_paragraph, with writer_paragraph being the reference category
- **(1 | proposition_id)** is a random effect to account for differences across propositions
- **(1 | writer_id)** is a random effect to account for individual writer differences
- **(1 | rater_id)** is a random effect to account for individual rater differences

The measurement scale of each rating attribute will determine the appropriate link function and the interpretation of the model coefficients, as follows:

For **continuous rating attributes** (e.g. paragraph relevance, writer confidence), which are all measured on a 0-100 scale, we will fit **mixed-effects beta regression models** using the logit link function, after transforming the 0-100 scale to the (0,1) interval by dividing by 100. The coefficients will be interpreted as:

- The coefficient for **paragraph_type = model_paragraph** represents the average difference in the logit-transformed rating attribute between LLM-written paragraphs and human-written paragraphs
- The coefficient for **paragraph_type = edited_paragraph** represents the average difference in the logit-transformed rating attribute between co-written paragraphs and human-written paragraphs

We will report coefficients with 95% confidence intervals. As a sensitivity analysis, we will compare results from beta regression with standard linear mixed-effects models on the original 0-100 scale.

For **nominal categorical rating attributes** (e.g. writer race, writer gender), we will fit **mixed-effects multinomial logistic regression model**s using the multinomial logit link function:

$$\log\left(\frac{P(Y=j)}{P(Y=\text{reference})}\right) = \beta_{0j} + \beta_{1j}X_1 + \beta_{2j}X_2 + \ldots + \beta_{pj}X_p + u_{0j}$$

for each category j (except the reference category), where u_{0j} represents the random effects. The coefficients represent the log-odds of being classified in category j versus the reference category when comparing different paragraph types. We will report these as odds ratios with 95% confidence intervals and use Cramer's V as the effect size measure.

For **ordinal categorical rating attributes** (e.g. writer education, writer income), we will fit **mixed-effects ordinal logistic regression models** using the cumulative logit link function:

$$\log\left(\frac{P(Y \leq j)}{P(Y > j)}\right) = \alpha_j - (\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p) + u_0$$

where α_j are category-specific threshold parameters and u_0 represents the random effects.

This model assumes proportional odds across categories. If this assumption is violated, we will fit a **partial proportional odds model** that allows some predictors to have category-specific effects:

$$\log\left(\frac{P(Y \leq j)}{P(Y > j)}\right) = \alpha_j - (\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \gamma_{1j} Z_1 + \gamma_{2j} Z_2 + \ldots) + u_0$$

We will report coefficients as odds ratios with 95% confidence intervals and measure Wilcoxon effect size (r).

In case of convergence issues, we will consider alternative model specifications with reduced random effects.

Alternative aggregation approach: As a sensitivity analysis, we will also conduct analyses using paragraph-level aggregated ratings (mean for continuous variables, mode for categorical variables) as the outcome variables, with paragraphs as the unit of analysis. This approach eliminates rater-level random effects whilst potentially reducing measurement noise. We will compare results between approaches to assess robustness.

Ecological validity subset analysis: As an additional sensitivity analysis to enhance the ecological validity of our findings, we will conduct the same mixed-effects regression analyses on a subset of the data that includes only writer-paragraph pairs where the writer expressed a preference for the LLM-assisted content (i.e. selected either "The paragraph on the right" or "Equal preference" in the paragraph preference question from Phase 1). This subset represents cases where LLM-assisted content would likely be published or shared in real-world scenarios, as writers who prefer their original human-written content would presumably not use the LLM-assisted version. We will compare effect sizes and significance levels between this subset and the full dataset to assess whether distortive effects are consistent across both samples.

5.2 RQ2: Variation in Third-Person Perceived Distortion

Our second main research question is concerned with establishing whether perceived distortion varies depending on which LLM is providing writing assistance, and what kind of inputs it receives from the writer. To answer this question, we will expand the **mixed-effects regression models** used for RQ1, specifying them as follows:

$$\begin{aligned}\text{attribute} \sim\ & \text{paragraph_type}+\\ & \text{paragraph_type:llm}+\\ & \text{paragraph_type:sub_condition}+\\ & (1 \mid \text{proposition_id})+\\ & (1 \mid \text{writer_id})+\\ & (1 \mid \text{rater_id})\end{aligned}$$

where:

- **paragraph_type:llm** is a conditional interaction that tests for effects of LLM assignment (n=3) on perceptions of model_ and edited_paragraph
- **paragraph_type:sub_condition** is a conditional interaction that tests for effects of input sub-condition assignment (n=3) on perceptions of model_ and edited_paragraph
- All other fixed and random effects are defined as in RQ1

The interaction terms allow us to examine whether the distortive effects observed in RQ1 vary systematically across different LLMs and input conditions.

For **continuous rating attributes**, we will again fit mixed-effects beta regression models using a logit link function, where the coefficients will be interpreted as:

- **paragraph_type:llm** coefficients: The additional difference in the logit-transformed rating attribute when using a specific LLM (compared to the reference LLM) for LLM-written or co-written paragraphs, beyond the main effect of paragraph type
- **paragraph_type:sub_condition** coefficients: The additional difference in the logit-transformed rating attribute when using a specific input condition (compared to the reference condition) for LLM-written or co-written paragraphs, beyond the main effect of paragraph type

For **categorical rating attributes** (both nominal and ordinal), we will fit the same expanded models using the appropriate link functions described in RQ1, with interaction terms interpreted in terms of odds ratios.

In case of convergence issues, we will consider alternative model specifications with reduced random effects.

5.3 RQ3: Writer Preference for LLM Writing

Our third, auxiliary research question examines to what extent writers prefer content written with LLM assistance to their own independent writing, and how this preference varies by choice of LLM and the type of input provided to the LLM.

In our analysis, we will use two measures of writer preference: 1) edit distance, as an implicit measure, and 2) stated preference, as an explicit measure.

5.3.1 Edit Distance

In Phase 1, for each proposition assigned to them, writers are asked to edit the LLM-written paragraph until it matches their opinion. If they feel it already matches their opinion, they do not have to edit it.

First, we will describe the distribution of edit distances across all writer-proposition pairs. Matching our approach for RQ1/RQ2, we will then fit a **linear mixed-effects regression model** specified as:

$$\begin{aligned}\text{edit_distance} \sim\ & \text{llm}+ \\ & \text{sub_condition}+ \\ & (1 \mid \text{proposition_id})+ \\ & (1 \mid \text{writer_id})\end{aligned}$$

where:

- **edit_distance** is a continuous measure of edit distance (e.g. Levenshtein distance) between the model_paragraph and edited_paragraph
- **llm** is a nominal categorical variable with 3 possible values for each of the 3 LLMs
- **sub_condition** is a nominal categorical variable with 3 possible values for each of the 3 input sub-conditions
- **(1 | proposition_id)** is a random effect to account for differences across propositions
- **(1 | writer_id)** is a random effect to account for individual writer differences

and interpretation as follows:

- The coefficient for each **llm** level represents the average difference in edit distance when using that LLM compared to the reference LLM, controlling for input condition and random effects.
- The coefficient for each **sub_condition** level represents the average difference in edit distance when using that input condition compared to the reference input condition, controlling for LLM choice and random effects.

In case of convergence issues, we will consider alternative model specifications with reduced random effects.

5.3.2 Stated Preference

After editing the LLM-written paragraph, writers in Phase 1 have to choose between the paragraph written with LLM assistance and the paragraph they wrote independently, for the purposes of communicating their opinion on the proposition. Writers can choose either paragraph or express equal preference.

To analyse which factors predict writer preferences, we will use a **logistic regression model**, with a binarised outcome variable. We will report results for two binarisation schemes: 1) where we collapse the equal preference label with preference for the co-written paragraph, and 2) where we collapse the equal preference label with preference for the human-written paragraph. We will specify the model as:

$$
\begin{aligned}
\text{is_preferred} \sim\ & \text{llm}+ \\
& \text{sub_condition}+ \\
& \text{edit_distance}+ \\
& (1 \mid \text{proposition_id})+ \\
& (1 \mid \text{writer_id})
\end{aligned}
$$

where all terms are defined as in the edit distance model, and the link function is a logit function.

5.4 RQ4: Writer Preference for Distortion

Our fourth, auxiliary research question examines the extent to which the distortive effects of LLM writing assistance align with the preferences for distortion expressed by the writers using the LLMs.

In Phase 1, we will ask all writers about which types of distortion from using AI writing assistance they would find desirable or undesirable. In our analysis, we will then compare these stated preferences to the actual distortion that we measure for RQ1 and RQ2. This will primarily be a descriptive analysis.

Writer preferences for distortion will have been measured on a 0-100 scale for all rating attributes. For continuous and ordinal categorical attributes, we will have asked about preferences regarding each direction individually (e.g. opinion about AI writing assistance making the writer sound *more* vs. *less* formal in their writing). For nominal categorical attributes, we will have asked a single question each (e.g opinion about making the writer seem like their gender is *different* than what people would assume based on their own independent writing).

First, we will produce a ranking of the types of distortion most desirable and least desirable to writers. For each type of distortion, we will use a **one-sample t-test** test to establish whether the average desirability rating across writers is different to 50, i.e. neutral preference, on the 0-100 scale. Second, we will compare this ranking to the distortion observed in our experiments, in terms of direction, size, and significance. Note that, the size of the observed distortion is measured on different scales for different types of rating attributes (e.g. Cohen's d for continuous attributes), so that a unified ranking of observed distortion across attributes cannot be produced, and comparisons will have to be made within groups of attributes (e.g. all continuous rating attributes).

5.5 Consideration of Control Variables

In several of our analyses described above, we will control for proposition-level, writer-level, and rater-level heterogeneity through the inclusion of random effects.

5.6 Software and Packages

Regression models will be fitted in R. All other analyses will be conducted in Python.

5.7 Transformations

1. We will transform the "writer agreement" variable (0-100 scale), which can only be interpreted relative to a given proposition, into a variable called "writer polarity" (0-100 scale), where $\text{writer_polarity} = |50 - \text{writer_agreement}| \cdot 2$. This polarity variable can be interpreted uniformly across all propositions as the extent to which the writer's position is perceived to be moderate (50 on the original agreement scale) or extreme (0 or 100 on the original agreement scale).
2. We will sort responses for the numeric "writer age" variable into bins [18-29, 30-39, 40-49, 50-59, 60-69, 70+] for consistency with how we record and analyse all other sociodemographic variables.
3. For edit distance measures, we will normalise by text length to account for varying paragraph lengths.
4. For categorical variables with more than two levels (e.g. education, income), we will use appropriate contrast coding.

5.8 Inference Criteria

Our inference criteria are the p-values and corresponding uncertainty intervals on our quantities of interest. We will use a p-value threshold of .05 for statistical significance. For multiple comparisons, we will apply Holm-Bonferroni correction within each group of rating attributes a) to f), as listed in §2.3.3.

5.9 Data Exclusion

We will exclude data from all participants who did not complete the study. This will be done prior to any data analysis to ensure data quality while maintaining the integrity of our random assignment.

5.10 Missing Data

We will employ list-wise deletion to address missing data for all models. In other words, we will exclude subjects from a given analysis if they have missing data for any of the variables included in that analysis.

5.11 Exploratory Analysis

Beyond our pre-registered analyses, we plan to conduct the following exploratory analyses:

- **Proposition-level heterogeneity:** We will explore how distortive effects vary by proposition characteristics, such as topic area, polarisation level, and average writer stance.
- **Writer-level moderators:** We will explore whether distortive effects vary by writer attributes, such as political ideology, knowledge, or confidence.
- **LLM-input interaction effects**: We will explore whether certain combinations of LLMs and input conditions produce particularly pronounced distortive effects by testing three-way interactions (paragraph_type:llm:sub_condition).
- **Linguistic analysis:** We will conduct linguistic analyses comparing human-written, LLM-written, and co-written paragraphs on dimensions such as sentiment, reading level, length, and grammaticality.

The results of these exploratory analyses will be clearly distinguished from our pre-registered analyses in our reporting.

References

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